

Chapter 3

Matrices

3.1 Introduction

This chapter sets out the essential facts about matrices that you have to know as well as you know your route from LSE to home if you are to survive your econometrics course. The starting point is the definition of the basic operations of addition, subtraction and multiplication for matrices. These are the foundations of matrix algebra. I then give the rules of matrix algebra that follow from these definitions. Finally I discuss the circumstances in which a matrix has an inverse, introducing the ideas of linear independence, spanning and rank.

The treatment in this chapter is very cookbook - intuition, not proof. It's intention is to give you quickly the essentials of what you need to know on matrices. But the rules for matrices, in particular matrix multiplication, can seem very arbitrary. I explain why this is not arbitrary in section 3.14 at the end of this chapter. There is no need for you to read this, look at it only if you are interested. I explain that matrices are a natural representation of linear functions, that matrix multiplication is implied by the composition of functions, and the results on the existence of a matrix inverse are implied by the general conditions for the existence of a function inverse. .

3.2 What is a Matrix?

A matrix is a rectangular array of numbers, for example the matrix A is

$$A = \begin{pmatrix} A_{11} & A_{12} & \dots & A_{1m} \\ A_{21} & A_{22} & \dots & A_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nm} \end{pmatrix}.$$

A has n rows and m columns, and is described as an $n \times m$ matrix. A_{ij} is the element in row i and column j . Assume that all the numbers I mention are real unless I explicitly say that you should consider the possibility that they are complex. (This arises only in the chapter on eigenvalues and eigenvectors.)

3.3 Definitions for Matrix Algebra

You can do algebra with matrices just as you can do algebra with numbers. The first stage in developing matrix algebra is to define addition, subtraction and multiplication with matrices. I am going to ignore division for now, and come back to it later in the form of the matrix inverse A^{-1} . The most important difference between algebra with matrices and algebra with the numbers is that you can do addition, subtraction and multiplication with any pair of numbers. This is not so with matrices so the definitions of these operations have to include a statement on which matrices the operations can be done with.

Addition and subtraction are straightforward.

Definition 3 (Matrix Addition) *Two matrices A and B can be added if they are both $n \times m$ matrices. The sum $A + B$ is defined as the $n \times m$ matrix for which component ij is $A_{ij} + B_{ij}$.*

Definition 4 (Matrix Subtraction) *Two matrix B can be subtracted from the matrix A if both A and B are $n \times m$ matrices. The difference $A - B$ is defined as the $n \times m$ matrix for which component ij is $A_{ij} - B_{ij}$.*

There are two types of multiplication to be considered. The first is multiplication of a matrix by a number, often referred to as a scalar in this context.

Definition 5 (Multiplication of a Matrix by a Number) *Any $n \times m$ matrix A can be multiplied by any number θ . The product θA is defined as the $n \times m$ matrix for which component ij is θA_{ij} .*

It is also possible to multiply some pairs of matrices.

Definition 6 (Multiplication of Two Matrices) *If B is an $n \times m$ matrix and C is a $m \times p$ matrix the product BC is defined as the $n \times p$ matrix for which component ij is*

$$\sum_{h=1}^m B_{ih} C_{hj}.$$

3.4 Matrix Algebra 1: When Matrix Algebra is

Like Ordinary Algebra

Some of the rules for matrix algebra are the same as the rules for ordinary algebra where the variables are real numbers. These matrix algebra rules follow straight from the definition of matrix addition and multiplication and the corresponding properties of ordinary addition and multiplication.

1. Commutative Addition

If A and B are $n \times m$ matrices

$$A + B = B + A.$$

2. Associative Addition

If A, B and C are all $n \times m$ matrices

$$(A + B) + C = A + (B + C).$$

so it is possible to write simply $A + B + C$.

3. Associative Multiplication of a Matrix by a Scalar

If λ and μ are numbers and A is a matrix

$$\lambda(\mu A) = (\lambda\mu)A$$

4. Distributive Scalar Addition and Multiplication

If λ and μ are numbers and A is a matrix

$$(\lambda + \mu)A = \lambda A + \mu A.$$

5. Distributive Matrix Addition and Scalar Multiplication

If λ is a number and A and B are $n \times m$ matrices

$$\lambda(A + B) = \lambda A + \lambda B.$$

6. Associative Multiplication

If A is an $n \times m$ matrix, B a $m \times p$ matrix and C an $p \times q$ matrix

$$(AB)C = A(BC).$$

so it is possible to write simply ABC .

7. Distributive Addition and Multiplication

If A is an $n \times m$ matrix, B and C $m \times p$ matrices and D an $p \times q$ matrix

$$\begin{aligned} A(B + C) &= AB + AC \\ (B + C)D &= BD + CD. \end{aligned}$$

8. Expanding Brackets

Distributive addition and multiplication imply that if A and B are $n \times m$ matrices and C and D $m \times p$ matrices

$$(A + B)(C + D) = AC + BC + AD + BD.$$

9. The Zero Matrix.

The zero matrix 0 has all elements 0 . It behaves like 0 in ordinary algebra.

$$\begin{aligned} A + 0 &= A \\ A0 &= 0. \end{aligned}$$

10. The Identity Matrix

The $n \times n$ identity matrix I^n has $I_{ii}^n = 1$ for $i = 1, 2, \dots, n$ and $I_{ij}^n = 0$ if $i \neq j$, so

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

In practice we omit the n , and use I for any identity matrix. The identity matrix behaves like 1 in ordinary algebra.

$$\begin{aligned} IA &= A \\ AI &= A. \end{aligned}$$

3.5 Matrix algebra 2: When Matrix Algebra is Not Like Ordinary Algebra

1. You cannot add many pairs of matrices. You can only add A and B if they have the same number of rows and columns so both A and B are $n \times m$ matrices.
2. You cannot multiply many pairs of matrices. You can only multiply A by B if the number of columns of A is the same as the number of rows of B , so A is a $n \times m$ matrix and B is $m \times p$ matrix.
3. Matrix multiplication is not commutative. There are some matrices A and B for which $AB = BA$, but in general $AB \neq BA$. For example

$$\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 3 & 1 \end{pmatrix}$$

but

$$\begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 3 & -2 \end{pmatrix}.$$

4. By definition the matrix A^{-1} is the inverse of A if

$$A^{-1}A = AA^{-1} = I.$$

Many matrices do not have an inverse. A matrix cannot have an inverse unless it is square, that is has the same number of rows and columns, and many square matrices do not have an inverse. See section 3.10 of these notes for details.

3.6 The Transpose of a Matrix

The transpose A' of the $n \times m$ matrix A is a $m \times n$ matrix with $A'_{ij} = A_{ji}$. You may see the notation A^T used for the transpose of A . The notation $\mathbf{x}'$ for the row vector with the same elements as the column vector $\mathbf{x}$ simply treats the n vector $\mathbf{x}$ as an $n \times 1$ matrix. Transposes are important in econometrics,

for example the ordinary least squares estimator is $(X'X)^{-1}X'y$ involves a transpose.

The definition of a transpose implies that

1. $(A + B)' = A' + B'$ because

$$(A + B)'_{ij} = (A + B)_{ji} = A_{ji} + B_{ji} = A'_{ij} + B'_{ij}.$$

2. $(A')' = A$ because $(A'_{ij})' = A_{ji} = A_{ij}$.
3. $I' = I$ because $I'_{ii} = I_{ii} = 1$ and if $j \neq i$ $0 = I_{ij} = I_{ji} = I'_{ij}$.
4. If A is an $n \times m$ matrix and B a $m \times p$ matrix

$$(AB)' = B'A'$$

$$\text{because } (AB)'_{ji} = (AB)_{ij} = \sum_{h=1}^m A_{ih}B_{hj} = \sum_{h=1}^m A'_{hi}B'_{jh} = \sum_{h=1}^m B'_{jh}A'_{hi} = (B'A')_{ji}.$$

This implies that if $A_1, A_2, \dots, A_{m-1}, A_m$ are matrices, and A_i an $n_{i-1} \times n_i$ matrix for $i = 1, 2, \dots, m$

$$(A_1 A_2 \dots A_{m-1} A_m)' = A'_m A'_{m-1} \dots A'_2 A'_1.$$

5. If the $n \times n$ matrix A has an inverse

$$(A')^{-1} = (A^{-1})'$$

because

$$I = (A^{-1}A) = I' = (A^{-1}A)' = A'(A^{-1})'$$

and

$$I = (AA^{-1}) = I' = (AA^{-1})' = (A^{-1})'A'.$$

so $A'(A^{-1})' = (A^{-1})'A' = I$ so the inverse of A' is $(A^{-1})'$.

6. If B is a 1×1 matrix, that is a scalar $B' = B$. Thus if x is an n vector, A an $n \times m$ matrix and y a m vector, $x' Ay$ is a scalar so

$$x' Ay = (x' Ay)' = y' A' x.$$

3.7 The Trace of a Square Matrix

If A is a square $n \times n$ matrix the trace of A , written as $tr A$, is the sum of the diagonal terms, that is

$$tr A = \sum_{i=1}^n A_{ii}.$$

Matrices that are not square do not have a trace. If B is an $n \times k$ matrix and C is a $k \times n$ matrix, BC is an $n \times n$ matrix and CB is a $k \times k$ so both BC and CB are square matrices. Further

$$\text{tr}(BC) = \text{tr}(CB).$$

This is easily proved by noting that

$$\text{tr}(BC) = \sum_{i=1}^n (BC)_{ii} = \sum_{i=1}^n \left(\sum_{j=1}^k B_{ij} C_{ji} \right) = \sum_{j=1}^k \left(\sum_{i=1}^n C_{ji} B_{ij} \right) = \sum_{j=1}^k (CB)_{jj} = \text{tr}(CB).$$

Econometricians are particularly fond of the fact that if $\mathbf{x}$ is an n vector, and A an $n \times n$ matrix, then $\mathbf{x}'A\mathbf{x}$ is a scalar, that is a 1×1 matrix. Thus $\mathbf{x}'A\mathbf{x} = \text{tr}(\mathbf{x}'A\mathbf{x})$ which implies that

$$\mathbf{x}'A\mathbf{x} = \text{tr}(\mathbf{x}'A\mathbf{x}) = \text{tr}(A\mathbf{x}\mathbf{x}').$$

3.8 Special Square Matrices

There are a number of special matrices you need to know about.

- The matrix A is **diagonal** if $A_{ij} = 0$ for all $i \neq j$, so all the off diagonal elements are zero. For example the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

is diagonal, but the matrix

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 4 & 3 \end{pmatrix}$$

is not diagonal.

- The matrix A is **upper triangular** if $A_{ij} = 0$ for all $i < j$, so all the elements below the diagonal are zero. For example the matrix

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix}$$

is upper triangular, but the matrix

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 2 \\ 0 & 1 & 3 \end{pmatrix}$$

is not upper triangular.

- The matrix A is **lower triangular** if $A_{ij} = 0$ for all $i > j$, so all the elements above the diagonal are zero. For example the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 3 & 2 & 0 \\ 4 & 0 & 3 \end{pmatrix}$$

is lower triangular, but the matrix

$$\begin{pmatrix} 1 & 0 & 2 \\ 3 & 2 & 0 \\ 4 & 0 & 3 \end{pmatrix}$$

is not lower triangular.

- The matrix A is **symmetric** if $A' = A$ that is $A_{ij} = A_{ji}$ for all i and j . For example the matrix

$$\begin{pmatrix} 1 & 3 & 4 \\ 3 & 2 & 0 \\ 4 & 0 & 3 \end{pmatrix}$$

is symmetric, but the matrix

$$\begin{pmatrix} 1 & 0 & 2 \\ 3 & 2 & 0 \\ 4 & 0 & 3 \end{pmatrix}$$

is not symmetric.

- The matrix A is **orthonormal** if A' is the inverse of A so $A A' = I$. For example the matrix

$$\begin{pmatrix} 2^{-\frac{1}{2}} & -2^{-\frac{1}{2}} \\ 2^{-\frac{1}{2}} & 2^{-\frac{1}{2}} \end{pmatrix}$$

is orthonormal because

$$\begin{pmatrix} 2^{-\frac{1}{2}} & 2^{-\frac{1}{2}} \\ -2^{-\frac{1}{2}} & 2^{-\frac{1}{2}} \end{pmatrix} \begin{pmatrix} 2^{-\frac{1}{2}} & -2^{-\frac{1}{2}} \\ 2^{-\frac{1}{2}} & 2^{-\frac{1}{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Using notation A^h for column h of the matrix A , and thinking of A^h as a vector the matrix A is orthonormal if $\|A^h\| = 1$ for all h , and $(A^h)' A^k = 0$ for $h \neq k$. This says that the vectors that make up the columns of A all have length 1 and are orthogonal to each other.

- The matrix A is **idempotent** if $A^2 = A$. (The notation A^2 is used in the obvious way to mean AA .)

For example the matrix

$$\begin{pmatrix} 3 & 6 \\ -1 & -2 \end{pmatrix}$$

is idempotent because

$$\begin{pmatrix} 3 & 6 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 3 & 6 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ -1 & -2 \end{pmatrix}.$$

The most important example of an idempotent matrix is $X(X'X)^{-1}X'$.

In Mathematics for Microeconomics you will meet the following definitions.

- The symmetric $n \times n$ matrix A is **positive semidefinite** if $\mathbf{x}'A\mathbf{x} \geq 0$ for all n vectors $\mathbf{x}$.
- The symmetric $n \times n$ matrix A is **positive definite** if $\mathbf{x}'A\mathbf{x} \geq 0$ for all n vectors $\mathbf{x}$ and $\mathbf{x}'A\mathbf{x} = 0$ if and only if $\mathbf{x} = 0$.
- The symmetric $n \times n$ matrix A is **negative semidefinite** if $\mathbf{x}'A\mathbf{x} \leq 0$ for all n vectors $\mathbf{x}$.
- The symmetric $n \times n$ matrix A is **negative definite** if $\mathbf{x}'A\mathbf{x} \leq 0$ for all n vectors $\mathbf{x}$ and $\mathbf{x}'A\mathbf{x} = 0$ if and only if $\mathbf{x} = 0$.

3.9 Inverse Matrices

The matrix A has an inverse A^{-1} if and only if $A^{-1}A = I$ and $AA^{-1} = I$. A matrix that has an inverse is said to be invertible. The critical thing that you must remember are

1. If a matrix has different numbers of rows and columns it cannot have an inverse.
2. If a matrix has the same number of rows and columns it may not have an inverse.
3. A matrix has at most one inverse. To see that suppose that both B and C are inverses of A so $BA = I$ and $AC = I$, then

$$B = BI = B(AC) = (BA)C = IC = C.$$

4. The inverse of A^{-1} is A . This is implied by $A^{-1}A = I$ and $AA^{-1} = I$.
5. If A and B are both $n \times n$ invertible matrices the matrix AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

This is easily proved by noting that $(AB)(B^{-1}A^{-1}) = ABB^{-1}A^{-1} = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$, and similarly $B^{-1}A^{-1}AB = I$.

6. More generally a similar argument implies that if $A_1, A_2, \dots, A_{m-1}, A_m$ are invertible $n \times n$ matrices, the matrix $A_1A_2\dots A_{m-1}A_m$ is invertible and

$$(A_1A_2\dots A_{m-1}A_m)^{-1} = A_m^{-1}A_{m-1}^{-1}\dots A_2^{-1}A_1^{-1}.$$

3.10 When Does a Matrix Have an Inverse?

There are various ways of approaching the question of whether a matrix has an inverse. A way of getting at this is working with row echelon matrices. This emphasizes methods for finding the inverse of a matrix using pen and paper. It is useful to know how to find the inverse of a 2×2 matrix. But with

If $A_{11}A_{22} - A_{12}A_{21} = 0$ the matrix

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

has no inverse. If $A_{11}A_{22} - A_{12}A_{21} \neq 0$ the matrix has an inverse

$$A^{-1} = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}^{-1} = (A_{11}A_{22} - A_{12}A_{21})^{-1} \begin{pmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{pmatrix}$$

Think of this as swapping the diagonal elements changing the sign of the off diagonal elements, and dividing by the determinant of the matrix which is $A_{11}A_{22} - A_{12}A_{21}$.

For more general matrices if you are doing algebra you need to think about whether the matrix has an inverse, but if you are confident that the inverse exists just write A^{-1} . On no account should you write out A^{-1} component by component. If you need the numbers use a computer. Econometrics packages have a matrix inversion routine built into them.

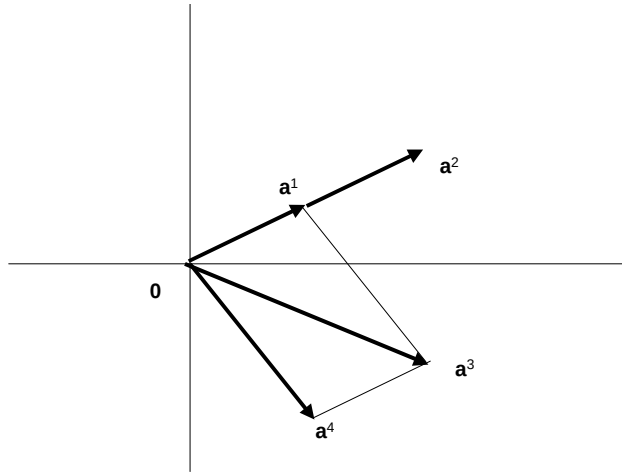
The big result on matrix inversion is:

Theorem 7 *Only square matrices have an inverse. Not all square matrices have an inverse. If A is a square $n \times n$ matrix either*

- *A has an inverse*
- *the only vector for which $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$*
- *the columns of A are linearly independent*
- *the columns of A span $\mathcal{R}^n$.*
- *rank $A = n$*
- *the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$ in $\mathcal{R}^n$*
- *the determinant $\det A \neq 0$*

or

- *A has no inverse*
- *there is a vector $\mathbf{x} \neq \mathbf{0}$ for which $A\mathbf{x} = \mathbf{0}$*
- *the columns of A are not linearly independent*
- *the columns of A do not span $\mathcal{R}^n$.*
- *rank $A < n$*
- *depending on $\mathbf{b}$ the equation $A\mathbf{x} = \mathbf{b}$ either has no solution or an infinite number of solutions*
- *the determinant $\det A = 0$.*

Figure 3.1: Linear Independence and Spanning in $\mathcal{R}^2$.

If any one of the first set of statements hold they all hold and none of the second set holds. Similarly if any one of the second set of statements holds all the second set and none of the first set hold. Thus any one of the first set of statements implies all the others, and any one of the second set of statements implies all the others.

I am about to explain what the term "linearly independent" means, I will then introduce spanning, and then explain rank. Determinants are defined in the chapter on determinants.

I do not prove theorem 7 here. But I do indicate in section 3.14 how a proof can be done.

3.11 Linear Independence

Definition 8 The set of vectors $\{\mathbf{a}^1, \mathbf{a}^2, \dots, \mathbf{a}^m\}$ is **linearly independent** if the only set of numbers $\{x_1, x_2, \dots, x_m\}$ for which

$$\mathbf{a}^1 x_1 + \mathbf{a}^2 x_2 + \dots + \mathbf{a}^m x_m = \mathbf{0}$$

is $x_1 = x_2 = \dots x_m = 0$.

You may find it helpful to look at Figure 3.1 at this point. In this Figure

- The vectors $\mathbf{a}^1$ and $\mathbf{a}^2$ are not linearly independent because $2\mathbf{a}^1 - \mathbf{a}^2 = \mathbf{0}$. Geometrically $\mathbf{a}^2 = 2\mathbf{a}^1$ so both vectors lie in the same straight line.
- The three vectors $\mathbf{a}^1$, $\mathbf{a}^4$ and $\mathbf{a}^3$ are not linearly independent because $\mathbf{a}^1 + \mathbf{a}^4 - \mathbf{a}^3 = \mathbf{0}$.
- Any two of $\mathbf{a}^1$, $\mathbf{a}^3$ and $\mathbf{a}^4$ are linearly independent.

It is in fact true, but I am not going to prove it, that no set of more than 2 vectors in $\mathcal{R}^2$ can be linearly independent, and much more generally that no set of more than n vectors in $\mathcal{R}^n$ can be linearly independent.

The idea of linear independence can be applied to the columns $\{A^1, A^2, \dots, A^m\}$ of the $n \times m$ matrix A . These are linearly independent if there is no set of numbers $x_1, x_2, \dots, x_m$ such that $A^1x_1 + A^2x_2 + \dots + A^mx_m = 0$. In matrix notation $A^1x_1 + A^2x_2 + \dots + A^mx_m = A\mathbf{x}$, so I can define:

Definition 9 *The columns of the matrix A are **linearly independent** if the only vector $\mathbf{x}$ for which $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.*

Linear independence matters for thinking about matrix inverses and solutions to simultaneous equations because of the following result:

Proposition 10 *The equation $A\mathbf{x} = \mathbf{b}$ has at most one solution if the columns of the matrix A are linearly independent. If the columns of A are not linearly independent, and the equation $A\mathbf{x} = \mathbf{b}$ has one solution, then the equation $A\mathbf{x} = \mathbf{b}$ has an infinite number of solutions.*

Proof. Suppose $\mathbf{x}_1$ and $\mathbf{x}_2$ are both solutions of the equation $A\mathbf{x} = \mathbf{b}$ so $A\mathbf{x}_1 = A\mathbf{x}_2$. Then $A\mathbf{x}_1 - A\mathbf{x}_2 = A(\mathbf{x}_1 - \mathbf{x}_2) = \mathbf{0}$. If the columns of A are linearly independent the only vector for which $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$ which implies that $\mathbf{x}_1 = \mathbf{x}_2$ so there is at most one solution to $A\mathbf{x} = \mathbf{b}$.

On the other hand suppose that there is a vector $\mathbf{x}_0 \neq \mathbf{0}$ with $A\mathbf{x}_0 = \mathbf{0}$. Then if $\mathbf{x}_1$ solves the equation $A\mathbf{x} = \mathbf{b}$, for any number λ $A(\mathbf{x}_1 + \lambda\mathbf{x}_0) = A\mathbf{x}_1 + \lambda A\mathbf{x}_0 = A\mathbf{x}_1 = \mathbf{b}$ because $A\mathbf{x}_0 = \mathbf{0}$. As λ can take an infinite number of different values there are an infinite number of different solutions. ■

3.12 Spanning

Definition 11 *The set of vectors $\{A^1, A^2, \dots, A^m\}$ **spans** $\mathcal{R}^n$ if for any $\mathbf{b}$ in $\mathcal{R}^n$ there are m numbers $x_1, x_2, \dots, x_m$ such that*

$$A^1x_1 + A^2x_2 + \dots + A^mx_m = \mathbf{b}$$

or in matrix notation

$$A\mathbf{x} = \mathbf{b}.$$

In words this says that the columns of A span $\mathcal{R}^n$ if the equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for every $\mathbf{b}$ in $\mathcal{R}^n$, and if the columns of A do not span $\mathcal{R}^n$ there are values of $\mathbf{b}$ for which the equation $A\mathbf{x} = \mathbf{b}$ has no solution. Taken together Proposition 10 and Definition 11 establish.

Proposition 12 • *If the columns of the $n \times m$ matrix A are linearly independent and span $\mathcal{R}^n$ then for every $\mathbf{b}$ in $\mathcal{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has exactly one solution.*

- If the columns of A are linearly independent but do not span $\mathcal{R}^n$, depending on $\mathbf{b}$ either
 - the equation $A\mathbf{x} = \mathbf{b}$ has one solution or
 - the equation $A\mathbf{x} = \mathbf{b}$ has no solution.
- If the columns of A are not linearly independent but do span $\mathcal{R}^n$ for every $\mathbf{b}$ in $\mathcal{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has an infinite number of solutions.
- If the columns of A are not linearly independent and do not span $\mathcal{R}^n$, depending on $\mathbf{b}$ either
 - the equation $A\mathbf{x} = \mathbf{b}$ has an infinite number of solutions or
 - the equation $A\mathbf{x} = \mathbf{b}$ has no solution.

3.13 Rank

3.13.1 Definition

The column rank of the matrix A is defined as follows:

Definition 13 The **column rank** of A is the integer r with the property that there is a set of linearly independent columns of A containing r elements, but there is no set of linearly independent columns of A containing more than r elements.

Thus r is the largest number of linearly independent columns of A . The row rank of the matrix A is defined as follows:

Definition 14 The **row rank** of A is the integer r with the property that there is a set of linearly independent rows of A containing r elements, but there is no set of linearly independent rows of A containing more than r elements.

In fact row and column rank are very simply linked

Proposition 15 The row rank and column rank of any matrix are the same.

This would be proved in a more thorough treatment of matrices. For this reason I can drop the terms "row rank" and "column rank" and simply use the term "rank".

If A is an $n \times m$ matrix with rank r the definitions and this result imply that $r \leq m$ and $r \leq n$. The definition of column rank implies:

Proposition 16 If A is an $n \times m$ matrix with rank r then the columns of A are linearly independent if $r = m$ and are not linearly independent if $r < m$.

You may find it helpful to look again at Figure 3.1. Think of the vectors in the figure as the columns of a 2×2 matrix.

- No single vector spans $\mathcal{R}^2$.
- The vectors $\mathbf{a}^1$ and $\mathbf{a}^2$ do not span $\mathcal{R}^2$.
- The vectors $\mathbf{a}^1$, $\mathbf{a}^3$ and $\mathbf{a}^4$ span $\mathcal{R}^2$, as do any two of these vectors.

The other important result about rank is:

Proposition 17 *If A is an $n \times m$ matrix with rank r then $r \leq n$, the columns of A span $\mathcal{R}^n$ if $r = n$ and do not span $\mathcal{R}^n$ if $r < n$.*

This is not obvious and I do not prove it. Thinking about this result with a two by two matrix, if the columns are $\mathbf{a}^1$ and $\mathbf{a}^2$ in Figure 3.1 these are not linearly independent, so $\text{rank } A < 2$ and the columns of A do not span $\mathcal{R}^2$. On the other hand if the columns of A are $\mathbf{a}^1$ and $\mathbf{a}^3$ these are linearly independent and also span $\mathcal{R}^2$.

3.13.2 Finding the Rank of a Matrix

You can investigate whether the m columns of the $n \times m$ matrix A are linearly independent by looking for nonzero solutions to $A\mathbf{x} = 0$. You could equally follow the same procedure with the rows of A , thinking about $\mathbf{z}A = 0$. It is usually quicker to use rows if $n < m$ and columns if $m < n$.

When working with 2 vectors it is worth remembering that they are linearly independent unless one or both are zero or one is scalar multiple of the other. For example the vectors $(1, 2, 3)$ and $(3, 2, 1)$ are linearly independent so the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}$$

has rank 2.

The vectors $(1, 2, 3)$ and $(4, 8, 12)$ are not linearly independent because

$$4(1, 2, 3) - (4, 8, 12) = 0$$

so the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 12 \end{bmatrix}$$

has rank 1.

For larger m and n there is a systematic way of finding the rank by finding something called the "row echelon" form of the matrix. Alternatively, if you have numerical values for the components of A you can ask your computer.

3.13.3 Rank and Solutions to Simultaneous Equations

Taken together Propositions 12, 16 and 17 imply:

Proposition 18 *If A is an $n \times m$ matrix, $\text{rank } A \leq n$ and $\text{rank } A \leq m$.*

- *If $\text{rank } A = m = n$ for every $\mathbf{b}$ in $\mathcal{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has exactly one solution.*

- If $\text{rank } A = m < n$ depending on $\mathbf{b}$ either
 - the equation $A\mathbf{x} = \mathbf{b}$ has one solution or
 - the equation $A\mathbf{x} = \mathbf{b}$ has no solution.
- If the $\text{rank } A = n < m$ for every $\mathbf{b}$ in $\mathcal{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has an infinite number of solutions.
- If $\text{rank } A < m$ and $\text{rank } A < n$ depending on $\mathbf{b}$ either
 - the equation $A\mathbf{x} = \mathbf{b}$ has an infinite number of solutions or
 - the equation $A\mathbf{x} = \mathbf{b}$ has no solution.

3.13.4 Rank and Inverse

I have already argued that if A has an inverse it is unique, so for every $\mathbf{b}$ in $\mathcal{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has exactly one solution $\mathbf{x} = A^{-1}\mathbf{b}$. It is also true (although I have not proved it) that the matrix A has an inverse if for every $\mathbf{b}$ in $\mathcal{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has exactly one solution. From the first part of proposition 18 this is only possible if A is a square $n \times n$ matrix. Taking into account the last part of proposition 18 gives the result on the matrix inverse

Proposition 19 *The square matrix $n \times n$ matrix A has an inverse if and only if $\text{rank } A = n$, a condition that can also be stated as*

- the columns of A are linearly independent, or equivalently
- the only vector for which $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.

These conditions are satisfied if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$ in $\mathcal{R}^n$.

3.14 A more sophisticated approach to matrices

The treatment in this chapter is very cookbook - intuition, not proof. It's intention is to give you quickly the essentials of what you need to know on matrices. But the rules for matrices, in particular matrix multiplication, can seem very arbitrary. This section sketches where they come from and how they relate to the existence of an inverse. This tells you what the steps in the proof are. You don't need to read it, but if you have a good mathematical background and are curious about how the argument works this indicates how it can be done.

- A function a from $\mathcal{R}^m$ into $\mathcal{R}^n$ is linear if for any $\mathbf{x}_1$ and $\mathbf{x}_2$ in and any real numbers γ_1 and γ_2

$$a(\gamma_1\mathbf{x}_1 + \gamma_2\mathbf{x}_2) = \gamma_1a(\mathbf{x}_1) + \gamma_2a(\mathbf{x}_2).$$

- A necessary and sufficient condition for $a : \mathcal{R}^m \rightarrow \mathcal{R}^n$ to be linear is that there is an $n \times m$ matrix such that

$$a(\mathbf{x}) = A\mathbf{x}$$

for all $\mathbf{x}$ in $\mathcal{R}^m$.

- Matrix multiplication comes from composition of functions. If $b : \mathcal{R}^n \rightarrow \mathcal{R}^p$ then

$$b(a(\mathbf{x})) = BA\mathbf{x}.$$

- If the function $a : \mathcal{R}^m \rightarrow \mathcal{R}^n$ has an inverse $a^{-1} : \mathcal{R}^n \rightarrow \mathcal{R}^m$ then a^{-1} is linear. The associated matrix is A^{-1} . If a does not have an inverse then A^{-1} does not exist.
- The function $a : \mathcal{R}^m \rightarrow \mathcal{R}^n$ has an inverse if it is one to one and onto. The function is onto if the columns of A span $\mathcal{R}^n$. It is one to one if the columns of A are linearly independent.
- If S_1 and S_2 are sets of vectors in $\mathcal{R}^n$ with p_1 and p_2 elements, the elements of S_1 are linearly independent, and the columns of S_2 span $\mathcal{R}^n$ then $p_1 \leq p_2$. As the columns of the $n \times n$ identity matrix are linearly independent and span this implies that

$$p_1 \leq n \leq p_2.$$

If $p_1 < n$ then the p_1 components of S_1 do not span. If $n < p_2$ the components of S_2 are not linearly independent. The members of a set of n elements of $\mathcal{R}^n$ is either linearly independent and spans or is not linearly independent and does not span.

- The dimension of a space is the minimum number of elements needed to span it.
- The rank of a matrix is the dimension of the space spanned by its columns.